Inventory Risk Decision System Report

Executive Summary

This project analyzes inventory data to identify parts at elevated risk of stockout and to support more proactive reorder decisions. The dataset contains approximately 500 parts and only 4 observed stockout events (0.8%), which makes traditional predictive machine learning unreliable for decision-making. Instead, the project emphasizes a transparent risk scoring framework that prioritizes operational action.

The strongest result is practical rather than purely statistical: the top 20% of highest-risk items capture 100% of observed stockouts, showing that the framework effectively concentrates attention on the most critical parts.

Key takeaways:

- Stockout events are extremely rare at 0.8%, so conventional model metrics can be unstable.
- The top 20% highest-risk items contain all observed stockouts in this dataset.
- A rule-based scoring model is more interpretable and operationally useful than forcing a fragile ML workflow.
- The output translates directly into action categories such as Immediate Reorder, Monitor Closely, Review, and Low Priority.

Business Problem

Inventory stockouts can delay production, increase expediting costs, and create lost-sales risk. The objective of this project is to use transactional inventory data to identify parts that deserve closer attention before a disruption occurs.

Methodology

The workflow begins with raw inventory master data and transaction history. Feature engineering in the notebook derives on-hand balance, average demand, days of supply, inventory value, demand volatility, and a simple stockout indicator. The project then combines two perspectives: lightweight machine learning exercise for exploration and a rule-based risk score for operational prioritization.

Risk Scoring Logic

Rule	Points
days_of_supply < lead_time_days	+3
lead_time_days > 20	+2
Daily_usage > median usage	+1

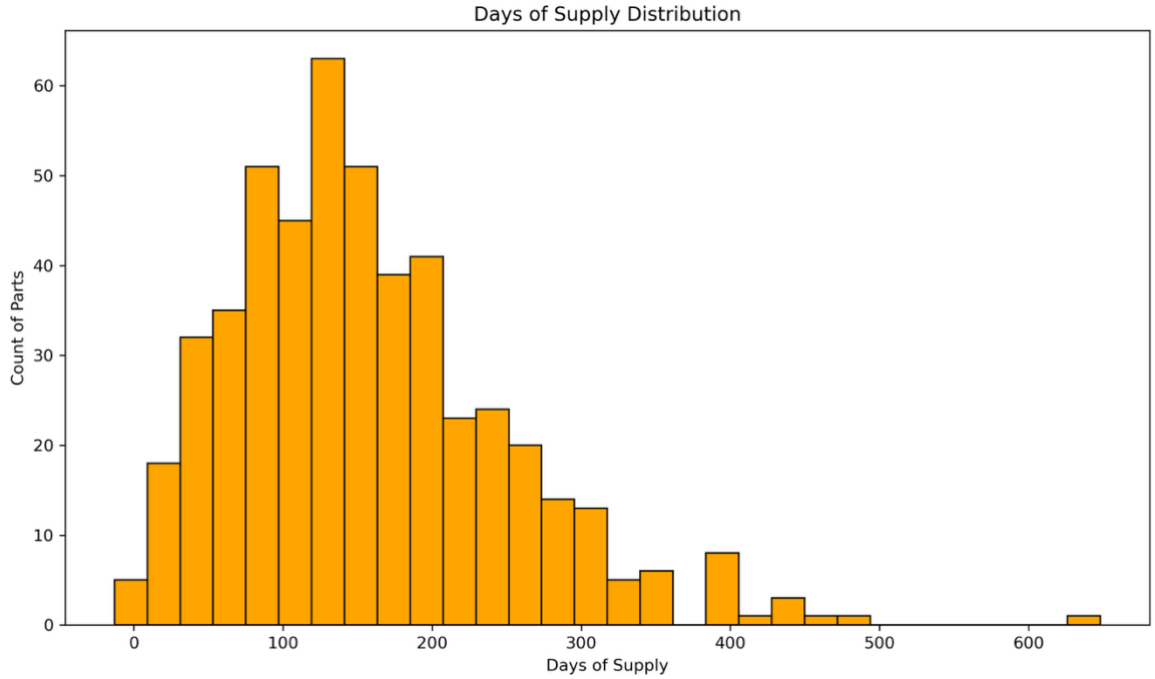
Estimated Business Impact

Using a simple assumption of $1,000 per stockout, preventing 5 avoidable stockouts per year would preserve roughly $5,000 in value. The more important operational benefit is prioritization: teams can focus limited planning capacity on the highest-risk slice of inventory instead of reacting uniformly to every part.

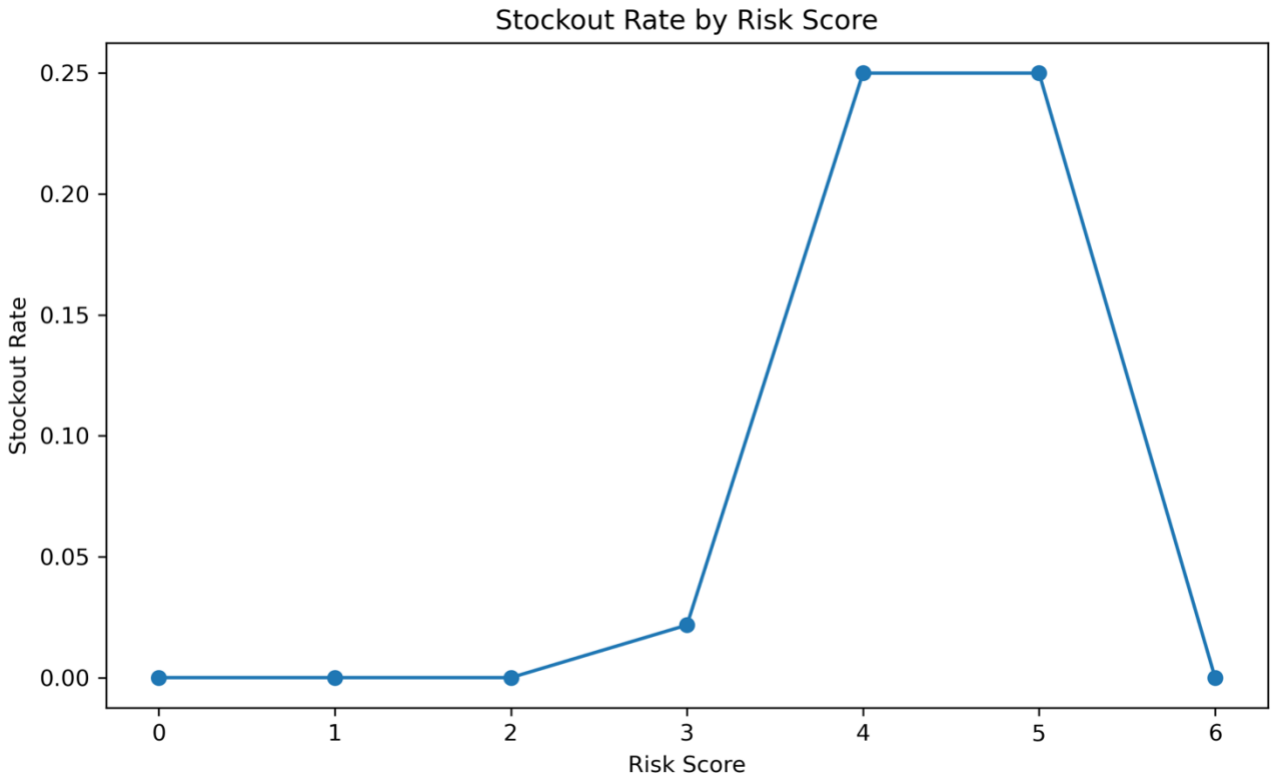
In this dataset, the top 20% of items contain 4 of 4 stockouts. That concentration corresponds to a 5.0x higher stockout rate than the overall portfolio baseline.

Inventory Patterns

The first two visuals show how the inventory base is distributed and how risk score relates to observed stockout behavior.

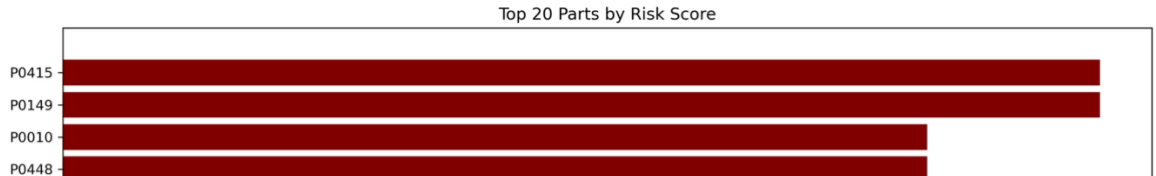


Days of supply is concentrated in a relatively narrow band for most parts, while a small tail of items exhibits very low coverage. Those low-coverage items become the natural candidates for elevated risk review.

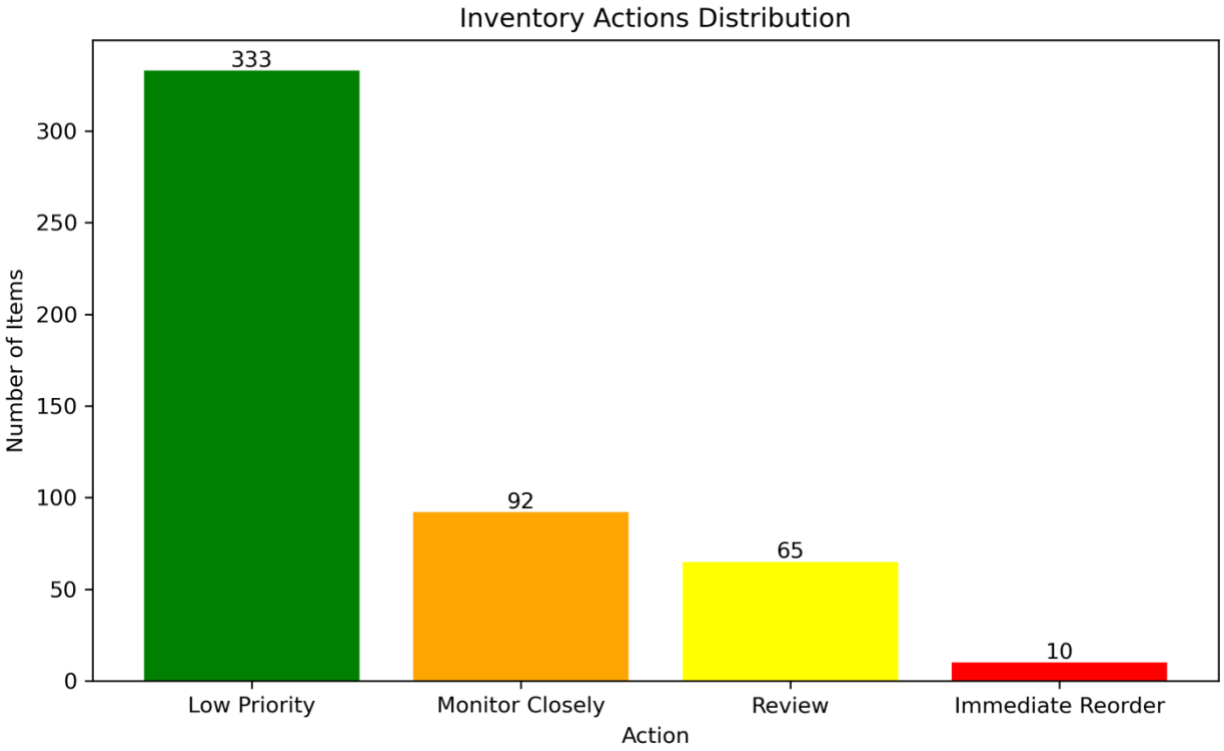


Stockout frequency rises sharply at the upper end of the risk score range, supporting the logic that the score is useful for sorting parts into decision-ready tiers.

Operational Prioritization Visuals

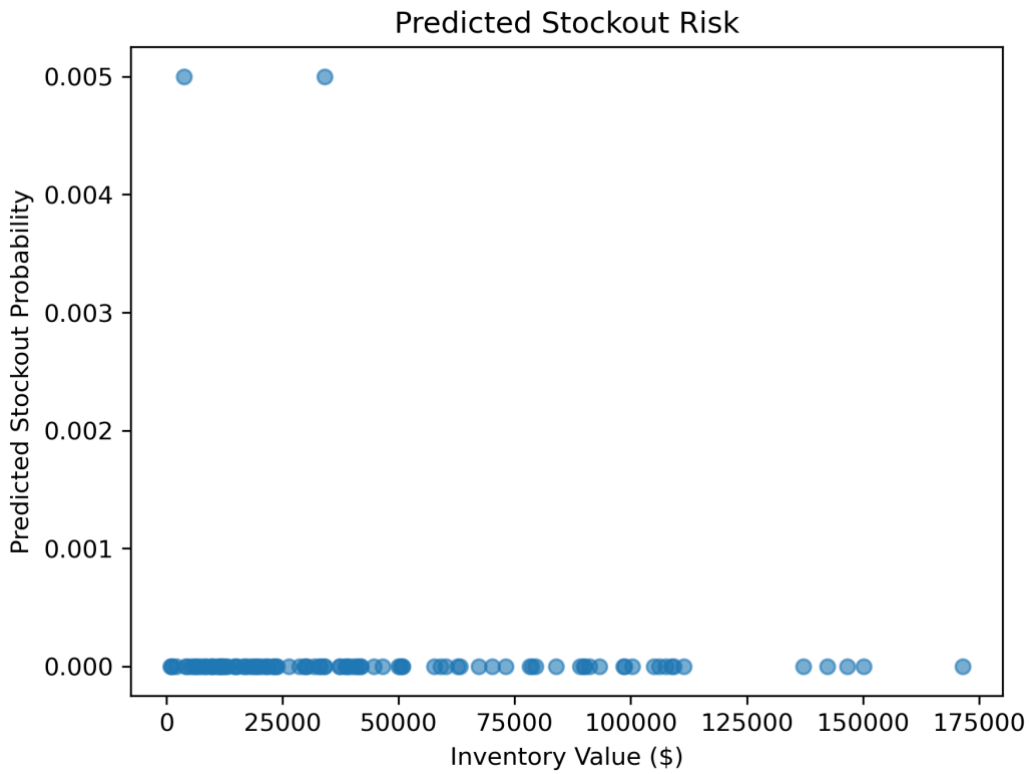


The highest-risk items cluster at the top of the ranking and provide a clear list for immediate review. This is the most actionable chart in the report because it maps directly into part-level intervention.

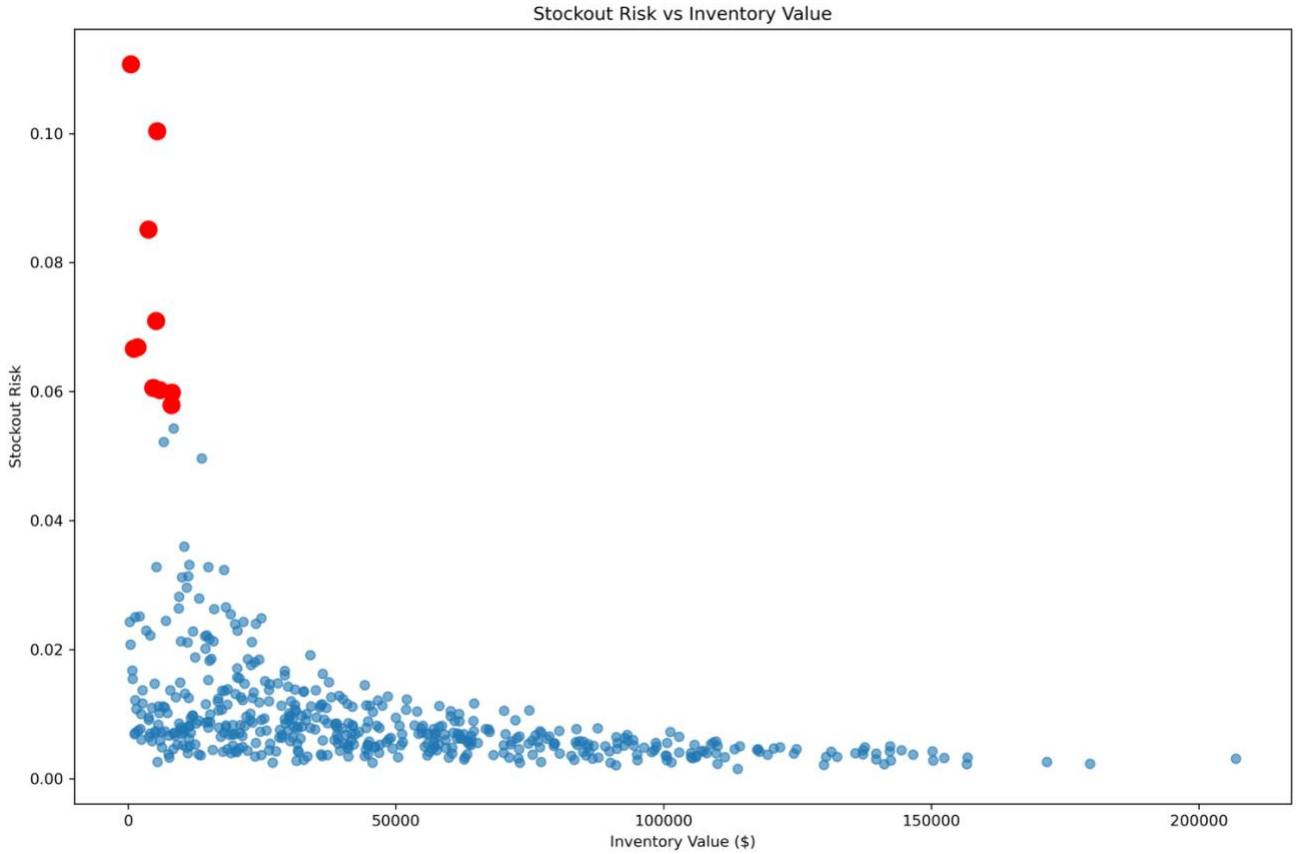


Most items remain low priority, while a small subset falls into Monitor Closely or Immediate Reorder categories. That is exactly the intended behavior of a prioritization framework.

Financial Risk Review



Items with both high inventory value and high predicted stockout probability represent the greatest financial risk and should be prioritized for immediate intervention.



This scatter plot shows how stockout risk is distributed across inventory value. It helps identify parts that are not only more likely to cause disruption but may also tie up meaningful dollars in inventory decisions.

Top 10 High-Risk Inventory Items

Part Number	Category	Supplier	Risk Score	Days of Supply	Action
P0149	Fasteners	MotionSupply	6	10.0	Immediate Reorder
P0415	Motors	ElectroParts	6	16.6	Immediate Reorder

Business Recommendations

- Prioritize review of the top 20% highest-risk items on a recurring planning cadence.
- Use the Immediate Reorder category to trigger fast intervention on parts with low coverage and long lead times.
- Monitor suppliers associated with repeatedly high-risk parts to determine whether lead-time reductions or alternate sourcing are warranted.
- Pair the risk score with reorder quantity recommendations so planners can move directly from diagnosis to action.

Limitations

- Only 4 stockout events are observed, so percentage-based validation metrics are highly sensitive to small count changes.
- The machine learning component is exploratory only and should not be over-interpreted as a production-ready classifier.

- The dataset is synthetic, so business value should be interpreted as a demonstration of decision logic rather than a live operational forecast.

Conclusion

This project demonstrates that an interpretable risk scoring framework can turn raw inventory transactions into a practical decision system. Rather than chasing unstable predictive metrics, it focuses on ranking, triage, and operational action - a framing that is more aligned with how inventory teams actually work.